

BIO DATA



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1. Educational Qualification

S.No	Branch / Specialization	Institute/University	Year of Passing
B.E.	Electrical and Electronics Engineering	Kamaraj College of Engineering and Technology, Virudhunagar	2006
M.E.	Power Electronics and Drives	Mepco Schlenk Engineering College, Sivakasi	2008
Ph.D.	Power Electronics	Anna University, Chennai	2024

2. Professional/ Industry Experience

: (in chronological order)

S.No	Designation	Institution/Organization	Period	
			From	To
1.	Assistant Professor	Ramco Institute of Technology, Rajapalayam.	10.05.2024	Till date
2.	Assistant Professor	Ramco Institute of Technology, Rajapalayam.	07.07.2014	30.04.2016

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3.	Senior Software Engineer	Larsen and Toubro Limited, Integrated Engineering Services, Mysore	04.08.2008	30.09.2011
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3. Research Interest: Power Converters, Power Quality, Renewable Energy, Electric Vehicle Battery Charging System

4. Publications

4.1 JOURNAL PUBLICATIONS

1. Jayanthi Kathiresan, **Narthana Sivaperumal**, Gnanavadivel Jothimani, Muralidharan Srinivasan, “**Single Switch Non-Isolated Quadratic High Gain Configuration For Renewable Energy Applications**”, REVUE ROUMAINE DES SCIENCES TECHNIQUES — SÉRIE ÉLECTROTECHNIQUE ET ÉNERGÉTIQUE, Vol. 71, no. 1, pp. 97-102, March 2026. DOI: <https://doi.org/10.59277/RRST-EE.2026.1.16>.
2. **Narthana Sivaperumal**, Jayanthi Kathiresan, Gnanavadivel Jothimani, “**A Transformer-Less High-Gain Cubic Converter With Reduced Components for Grid-Tied PV Systems**”, International Journal of Circuit Theory and Applications, Vol. 54, no. 3, pp. 1714-1736, March 2026. DOI: <https://doi.org/10.1002/cta.70061>.
3. **Narthana Sivaperumal**, Gnanavadivel Jothimani, Albert Alexander Stonier, Geno Peter, Vijayakumar Arun, Vivekananda Ganji, “**Enhanced one stage interleaved PFC Cuk configuration-based EV charger with reduced components**”, Computers and Electrical Engineering, 126, 110497, August 2025. DOI: <https://doi.org/10.1016/j.compeleceng.2025.110497>.
4. **Narthana Sivaperumal**, Gnanavadivel Jothimani, “**An energy efficient unidirectional on-board battery charger for power factor correction in electric vehicles**”, Electrical Engineering, Vol. 106, pp. 4127–4142, Jan 2024. DOI: <https://doi.org/10.1007/s00202-023-02208-4>.
5. **Narthana S**, Gnanavadivel J, “**Power Quality Analysis of Interleaved Cuk Configuration-Based Interval Type-2 Fuzzy Logic Controller for Battery Charging in Electric Vehicles**”, Arabian Journal for Science and Engineering, Vol. 49, no. 5, pp. 6259 – 6273, May 2024. DOI: <https://doi.org/10.1007/s13369-023-08189-7>.
6. **Narthana Sivaperumal**, Gnanavadivel Jothimani, “**A single-stage bridgeless isolated positive output Cuk configuration-based unidirectional onboard battery charger**”, International Journal of Circuit Theory and Applications, Vol. 52, no. 1, pp. 396-419, Jan 2024. DOI: <https://doi.org/10.1002/cta.3757>.
7. S. **Narthana**, P. Muthu Thiruvengadam & J. Gnanavadivel, “**Adaptive neuro-fuzzy approach for maximum power point tracking with high gain converter for photo voltaic applications**”, International Journal of Advanced Technology and Engineering Exploration, Vol: 9(93), pp. 1168 – 1182, August 2022. DOI: <http://dx.doi.org/10.19101/IJATEE.2021.875193>.
8. J.Gnanavadivel, M.Shunmathi, P.Muthu Thiruvengadam, S.Narthana, “**Analysis of DC-DC Converter with High Step -up Gain for Alternative Energy Sources**”,

4.2 CONFERENCE PUBLICATIONS

1. Narthana S, presented a paper titled “**Novel Harmonic Elimination Technique using Fuzzy Logic Controller in Sinusoidal Pulse Width Modulated Cascaded Multilevel Inverter**”, in the National Level Conference on Emerging Technologies in Electrical & Electronics Systems, Noorul Islam College of Engineering, Kumaracoil, Thuckalay during March 08, 2008.
2. Narthana S, presented a paper titled “**Novel Harmonic Elimination Method in Cascaded Multilevel Inverter using Fuzzy Logic**”, in the Second National Conference on Power Electronics and Power Systems – MEPPCON’08, Mepco Schlenk Engineering College, Sivakasi during March 14, 2008.
3. S. Narthana, presented a paper titled “**Design of High Gain DC-DC Converter with Active Switched Capacitor for Photo Voltaic Systems**”, in the International Virtual Conference on Power, Information and Communication (ICPIC), organized by PSN College of Engineering and Technology, Tirunelveli during May 12, 2021.
4. S. Narthana, presented a paper titled “**An IoT Based Transmission Line Power Monitoring System for Fault and Sag Detection**”, in the International Conference on Recent Trends in Engineering and Technology (ICRTET-25), held at Arasu Engineering College, Kumbakonam on May 14, 2025.
5. **S. Narthana**, presented a paper titled “**Medicine Dispenser**”, in the International Conference on Recent Trends in Engineering and Technology (ICRTET-25), held at Arasu Engineering College, Kumbakonam on May 14, 2025.
6. **S. Narthana**, presented a paper titled “**Medicine Dispenser**”, in the International Conference on Recent Trends in Engineering and Technology (ICRTET-25), held at Arasu Engineering College, Kumbakonam on May 14, 2025.
7. **Narthana S**, Arulkumar U, Sri Yogesh B A, Rahul K, presented a paper titled “A Sustainable Robotic Solution for Automated Solar Panel Maintenance”, in the 10th International Conference on Engineering Technology and Science (ICETS’ 2026), held at Muthayammal College of Engineering, Rasipuram on March 28, 2026.
8. **Narthana S**, Sivapragashpathy R, Subash K, Raghul S, presented a paper titled “**REAL-TIME MONITORING AND AUTOMATED BLOCK REMOVAL IN SEWAGE PIPELINES USING AN IOT-BASED ROBOTIC PLATFORM**”, in the Second International Conference in Pathway of Electrical, Automation, Communication and Electronics (PEACE-2026), held at K. Ramakrishnan College of Technology (Autonomous), Tiruchirappalli on March 28, 2026.

4.3 Book/Book chapter publications

Nil

5. Funded Projects

S.No	Title of the project	Funding Agency	Duration	Amount	Status
1.	AI-Driven Smart Assistive Navigation Stick for Empowering Visually Impaired Individuals	IUCEE Student Project Competition 2026	8 Months	Partial Financial Support	Selected in the First Screening Process
2.	SMART PLANT PHENOTYPING USING SPECTRAL-AI	MSME IDEA HACKATHON 5.0	1 year	15,00,000/-	Applied
3.	SMART VISION-BASED DRIVER ASSISTANCE WITH ADVANCED NOTIFICATION SYSTEM	AICTE RPS 2025 Lab to Market	3 years	48,00,000	Applied
4.	Low-Cost IoT-Integrated Sewage Cleaning Robot for Smart Sanitation	TNSCST	1 year	10,000	Applied
5.	SmartAgriSort: Affordable IoT-Powered Crop Grading and Storage System for Rural Agriculture	TNSCST	1 year	10,000	Applied
6.	AI-Enabled Spectral Plant Phenotyping for Sustainable Agriculture	RIT-SEED 2025	6 Months	50,000	Applied
7.	Low-Cost IoT-Integrated Sewage Cleaning Robot for Smart Sanitation	RIT-SEED 2025	6 Months	50,000	Applied

6. Ph.D. guidance

S.No	Scholar name	University	Topic	Status
Nil				

7. Short term Courses / Seminars /FDP/Workshop/Conference Attended:

S.No.	Name of Programme/event	Duration		Total number of days
		From	To	
1.	6 days offline ATAL FDP on “Cutting-Edge AI for Autonomous Vehicles: Advanced Research and Innovations”	09.12.2024	14.12.2024	6 days
2.	NPTEL-AICTE FDP on “Patent Search for Engineers and Lawyers”	Aug 2025	Oct 2025	8 weeks

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3.	6 days online short-term programme on “Sustainable Smart Energy Technologies”	06.01.2025	11.01.2025	6 days
4.	Naan Mudhalvan FDP on “PCB Design by using Fusion”	10.02.2025	16.02.2025	7 days
5.	Professional Development Programme on “Educational Video Production”	16.07.2025	21.07.2025	6 days
6.	Five days Faculty Development Programme (FDP) on Competency Based Engineering Education	10.02.2025	15.02.2025	6 days
7.	Two Days Workshop on “From Site Assessment to Simulation: A Hands-On Solar PV Design”	09.10.25	10.10.2025	2 days
8.	One-day online National Level Workshop on “Ethical Research and Publication using AI Tools”	01.11.2025	-	1 day
9.	ATAL FDP on “Next-Gen Smart Microgrids: Technologies & Challenges in Renewable Energy Integration”	01.12.2025	06.12.2025	6 days
10.	AI and Machine Learning for Electrical Engineering	05.05.2025	09.05.2025	5 days
11.	FDP on “Virtual Labs”	22.10.2024	24.10.2024	3 days
12.	One day workshop on “Exploring AI Tools for Effective Academics”	29.07.2024	-	1 day
13.	Short Term Course on NI LabVIEW Core 1 & Core 2	15.06.2015	19.06.2015	5 days
14.	Embedded Systems and Its Applications	21.12.2015	22.12.2015	2 days
15.	Soft Computing Techniques for Applications in Electrical Sciences	07.11.2014	08.11.2014	2 days
16.	Challenges in Grid Connected Renewable Energy Sources	25.06.2015	26.06.2015	2 days
17.	Research Opportunities, Challenges in Power Electronics for EV and Its Impact on Smart Grid	14.12.2020	20.12.2020	6 days
18.	NaMPET Sponsored “Emerging Technological Challenges in Electric Vehicle (EV) System”	21.12.2020	23.12.2020	2 days
19.	Modelling of Solar & Wind Energy System with Design Techniques of Converter and Inverter Topologies	12.03.2021	-	1 day
20.	AICTE - ISTE Sponsored Induction/Refresher programme “Solar PV System Design Using IOT” (Phase - II)	31.03.2021	08.04.2021	10 days
21.	SERB Core Research Grant Sponsored “Challenges and Opportunities in Future Electrical Vehicles Charging”	12.04.2021	14.04.2021	3 days

22.	AICTE - ISTE Sponsored Induction/Refresher programme “Electric Vehicle Evolution - Impact on Power Grid”	19.04.2021	24.04.2021	6 days
23.	Application of AI & Machine Learning in Electric Vehicle Technology	10.12.2021	11.12.2021	2 days
24.	Design Optimization of High Frequency Transformers and Inductors	27.12.2021	-	1 day
25.	Integration of Renewable Energy & EV to Microgrid: Prospects and Challenges	21.02.2022	25.02.2022	5 days
26.	Research Aspects in Power Electronic Converters for Microgrid System	26.05.2022	27.05.2022	2 days
27.	Two-Week Fulbright-Nehru Specialist Programme on “Research and Publication Ethics”	25.07.2022	06.08.2022	14 days
28.	‘Simulation of Power Converters using Matlab – Simulink’	01.08.2022	02.08.2022	2 days
29.	Thesis Writing and Patent - Filing	14.09.2022	-	1 day
30.	Artificial Neural Network Based Switching Loss Estimation for DC-DC Converters	02.11.2022	-	1 day
31.	3 Days Capacity Building Program on “Assembling and Servicing of Electric Vehicles”	15.11.2022	17.11.2022	3 days
32.	Maintenance and Troubleshooting of Electrical Appliances	18.01.2023	20.01.2023	3 days
33.	IP Awareness Training Program under National Intellectual Property Awareness Mission	09.02.2023	-	1 day
34.	Advanced Power Electronics and Storage Systems for Smart Grid	19.05.2023	20.05.2023	2 days
35.	Optimization Methodology in Power Electronics	22.06.2023	-	1 day
36.	Understanding the Basics of Temperature and Pressure Calibration	16.11.2023	-	1 day
37.	Sustainable Energy and Mobility: Prospects and Research Challenges in Power Electronics	08.12.2023	09.12.2023	2 days

8. Short term Courses / Seminars /FDP/Workshop/Conference organized:

Sl.No.	Name of Programme/event	Duration		Total number of days
		From	To	
1.	Seminar on Awareness Program in "Career	12.11.2024	-	Half-a-day

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	opportunities in Solar and Electric Vehicle"			
2.	Short-Term Training Programme on “Emerging Trends in Sustainable E-Mobility and Electric Power Train Technologies”	15.12.2025	20.12.2025	6 days
3.	Awareness Program on “PM Surya Ghar: Muft Bhijili Yojana – Solar PV Rooftop Subsidy	01.12.2025	Till date	-
4.	Value Added Course on “Selection and Testing of Industrial Sensors”	20.01.2025	25.01.2025	5 days
5.	Value Added Course on “Creative Electronics with Arduino for Practical Applications”	20.01.2025	22.01.2025	3 days
6.	Value Added Course on “Electronic Gadgets Design and Development Lifecycle”	30.06.2025	04.07.2025	5 days

9. Online Courses

S.NO	COURSE TITLE	UNIVERSITY	DATE
1.	Design of Power Electronic Converters	Indian Institute of Technology, Guwahati	19.02.2024 to 13.04.2024
2.	Differential Equations for Engineers	The Hong Kong University of Science and Technology	24.07.2024 to 06.11.2024
3.	Patent Drafting for Beginners	IIT Madras	20.01.2025 to 14.02.2025 (4 weeks)
4.	Patent Search for Engineers and Lawyers	IIT Kharagpur	18.08.2025 to 10.10.2025 (8 weeks)
5.	Roadmap for Patent Creation	IIT Kharagpur	19.01.2026 To 13.03.2026 (8 weeks)

10. Patents filed/awarded: -

SL. NO.	TITLE	APPLICATION NO.	FILED DATE	STATUS	INVENTOR(S)
1.	AN INTELLIGENT IOT-INTEGRATED MEDICATION DISPENSING SYSTEM FOR COMMUNITY HEALTHCARE ACCESS	202541107990	31.10.2025	Published	Mr. M. Harish Mr. A.B. Pranesh Mr. R. Darwin Srithar Dr. S. Kannan Dr. S. Narthana Ms. S. Sharmila Kumari
2.	A Compact Transformerless Interleaved Buck-Boost LED Driver	202341034005 A	15.05.2023	Published	Dr.J.Gnanavadivel Ms.P.Muthu Thiruvengadam Mrs.S.Narthana Dr.S.Muralidharan

11. Copyright filed/awarded:

SL. NO.	TITLE	APPLICATION NO.	FILED DATE	STATUS	INVENTOR(S)
1.	Embedded C Programming: A Hands-on Beginner's Guide	0290/2025-CO/L	20.03.2025	Granted	Mrs. S. Sharmila Kumari Mrs. G. Sivapriya Dr. S. Narthana Dr. S. Kannan

12. Membership in Professional societies:

- Associate Member in Institution of Engineers (India), Kolkata for life time - (Membership No. AM3190725)
- Member in Indian Society of Technical Education for life time. (Membership No. LM101168)

13. Industrial Training

SL. NO.	Name of the Industry	Duration	
		From	To
1.	Rajapalayam Mills Ltd, The Ramco Group Textile Division, Rajapalayam	16.07.2024	18.07.2024
2.	Gowri House Metal Works LLP, The Ramco Group, Rajapalayam	22.12.2025	24.12.2025

14. Courses Taught:

Academic Year	Semester	Course Code	Course Name	Pass Percentage
2024-2025 (Odd)	VII Sem. B.E. EEE	EE3701	High Voltage Engineering	60 / 64 = 93.75%
2024-2025 (Odd)	VII Sem. B.E. Civil	CME365	Renewable Energy Technologies	35 / 35 = 100%
2024-2025 (Even)	IV Sem. B.E. EEE	EE3401	Transmission & Distribution	56 / 65 = 86.15%
2014-2015 (Even)	IV Sem. B.E. EEE	EE6402	Transmission & Distribution	64 / 69 = 92.75%
2015-2016 (Even)	VI Sem. B.E. EEE	EE6601	Solid State Drives	67 / 69 = 97.1%
2015-2016 (Even)	IV Sem B.E. EEE	EE6402	Transmission & Distribution	69 / 72 = 95.83%

15. Awards, Recognition& Achievements: -

- **6 SCI** and **2 Scopus** journal publications
- **3 International** and **2 National conference** publications
- Produced 100 % result in below subject,
CME365 – Renewable Energy Technologies
- Received the Mentor Certificate for NPTEL course “Fundamentals of Electrical Engineering”
- Awarded for the Mentoring the team RIT_ECOLITES participated in IITM PALS innoWAH! 2024-25, conducted by IITM PALS.